Department of Computer Science
SUKKUR IBA UNIVERSITY
Kandhkot Campus
Machine Learning
Fall 2026 (Mid Term Lab Exam)

Instructor: Muhammad Ismail Qureshi **Date :19-10-2026**
Marks: 15 **Time: 150 minutes**

Name: **Enrollment No:**

Instructions for Students:

- Submit your completed handout.

1. Decision Tree Classifier Using the Iris Dataset:
    - Lab Objective
        1. To develop, train, evaluate, and visualize a Decision Tree Classifier using the built-in Iris dataset available in Scikit-learn.
    - Problem Statement
        1. You are required to build a machine learning model that classifies Iris flowers into one of three species based on four measurements:
            1. Sepal length (cm)
            2. Sepal width (cm)
            3. Petal length (cm)
            4. Petal width (cm)
        2. The target classes are Iris setosa, Iris versicolor, and Iris virginica.

Lab Tasks

Task	Description
1	Load and explore the Iris dataset
2	Separate features and target; split into training and testing sets
3	Build and train a Decision Tree Classifier
5	Visualize and interpret the decision tree
6	Predict the species of a new flower

4. Programming Requirements

Use Python with the following libraries:

- `scikit-learn`
- `pandas`
- `matplotlib`

Your implementation must satisfy these requirements:

1. Load the Iris dataset using `load_iris()`.
2. Display the first five records, feature names, target names, and dataset shape.
3. Train a `DecisionTreeClassifier` using the Gini criterion and `max_depth=3`.
4. Plot the trained decision tree with feature names and class names.

5. Predict the proabablities

Laboratory Exam Assessment Rubrics

Criteria	Poor performance (0-30%)	Average (30-60)%	Good (60-80)%	Exceptional (80-100)%	Obtained Score
Theoretical Calculations	Inadequate or incorrect theoretical calculations, significant errors, and a lack of understanding of relevant concepts	Basic understanding of theoretical calculations, but with several mistakes or omissions.	Accurate theoretical calculations with minor errors or omissions.	Highly accurate and thorough theoretical calculations, demonstrating a deep understanding of the underlying principles.	
Multisim Circuit	Incomplete or incorrect Multisim circuit, limited use of relevant components or connections, and poor simulation results.	Basic use of Multisim with some errors in the circuit design and simulation results.	Competent use of Multisim, with minor errors or improvements needed in the circuit design or simulation.	Highly effective use of Multisim, with an accurate and well-designed circuit that produces excellent simulation results.	
Results Analysis	Incomplete or incorrect analysis, poor understanding of data, and a lack of relevant conclusions.	Basic data analysis with some inaccuracies or incomplete conclusions.	Competent data analysis, with minor inaccuracies and reasonable conclusions drawn.	Thorough and highly accurate data analysis, with meaningful conclusions that demonstrate a deep understanding of the experiment.	
Results reporting	Inadequate or poorly organized report, significant errors, and a lack of clarity in presentation.	Basic report with some organization issues and minor errors in presentation.	Well-organized report with minor presentation issues.	Exceptionally well-structured and presented report, demonstrating a clear and concise communication of findings and methods.	